Lecture 6 – Simple Harmonic Motion (SHM)

- A/Prof Tay Seng Chuan
Email: pvotaysc@nus.edu.sg

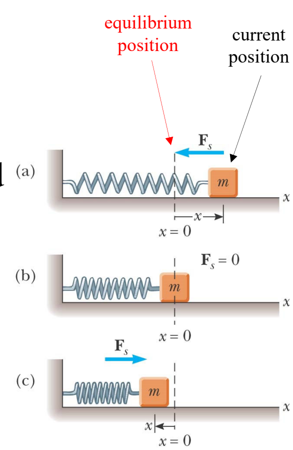
Grandfather clocks use SHM for accurate timekeeping, and musical instruments like guitars utilize SHM for sound vibration and pitch. More importantly, bridges use damped SHM to absorb shocks and prevent collapse. These examples show how SHM principles are applied in everyday objects and systems, and we will look at the details

A few slides in this chapter come from Chapter-14 of Young's textbook.
This PPT file will well define the scope of your tests.



# Periodic Motion

- ***Periodic motion*** is motion of an object that regularly repeats its previous pattern. The object returns to a given position after a fixed time interval
- A special kind of periodic motion occurs in mechanical systems when the force acting on the object is proportional to the position of the object relative to some equilibrium position, and if the force is always directed toward the equilibrium position, the motion is called ***simple harmonic motion*** (**SHM**)

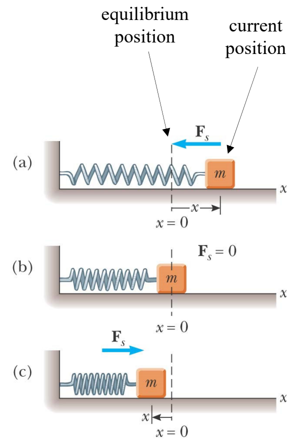


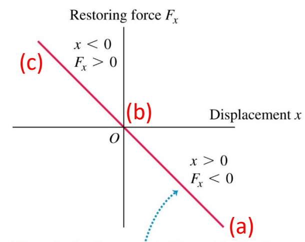
## Two Mandatory Conditions for the Force in Simple Harmonic Motion Described in Informal Language

(i) If the magnitude of actual position is further away from the equilibrium position on any sides, the magnitude of the force at that moment will be larger. Similarly, if the magnitude of the actual position is closer to the equilibrium position on any sides, the magnitude of the force at that moment will be smaller.

(ii) The direction of the force at anytime is in the opposite direction of the actual position at that time.

- A/Prof Tay



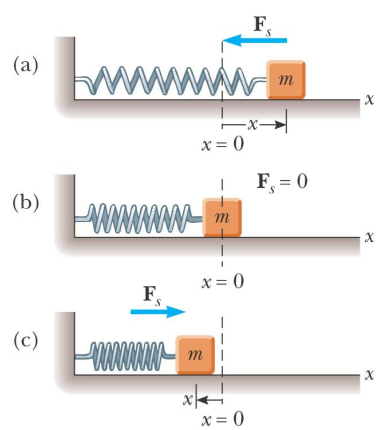

The restoring force exerted by an idealized spring is directly proportional to the displacement (Hooke's law, $F_x = -kx$): the graph of $F_x$ versus $x$ is a straight line.

3

## Motion of a Spring-Mass System

- A block of mass $m$ is attached to a spring, the block is free to move on a frictionless horizontal surface
- When the spring is neither stretched nor compressed, the block is at the ***equilibrium position***
  - ✓ $x = 0$

  The length of the spring at that moment is its natural length.



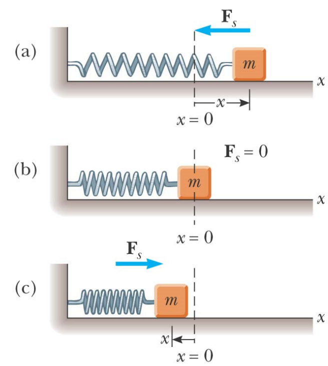

$x$ is measured from the end of the natural length of the spring. Treat going right as positive.

4

# Hooke's Law Re-visited

- Hooke's Law states

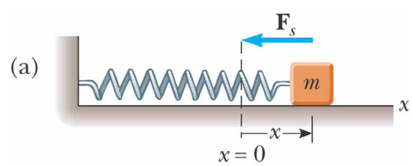
  Spring Force $F_s = - kx$
  - $F_s$ is the restoring force
    - ✓ It is always directed toward the equilibrium position
    - ✓ Therefore, it is always opposite to the displacement from equilibrium
  - $k$ is the force (spring) constant
  - $x$ is the displacement from the equilibrium position



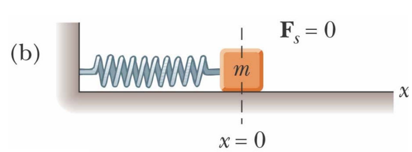

5

# More About Restoring Force $F_s = -kx$

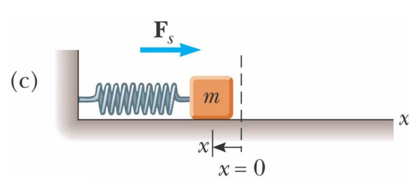
If the block is displaced to the right of $x = 0$
- The position is positive
- The restoring force is directed to the left




If the block is at the equilibrium position
- $x = 0$
- The spring is neither stretched nor compressed
- The restoring force is 0




If the block is displaced to the left of $x = 0$
- The position is negative
- The restoring force is directed to the right




6

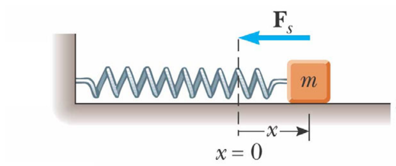
# Acceleration

- The force described by Hooke's Law is the net force in Newton's Second Law

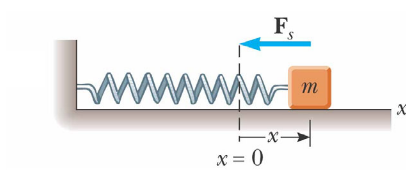
$$F_{\text{Hooke}} = F_{\text{Newton}}$$

$$-kx = ma_x$$

$$\rightarrow \quad a_x = -\frac{k}{m}x$$





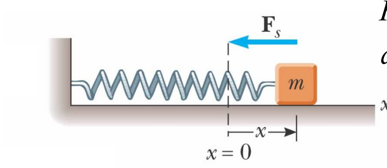




# Acceleration, cont.

$$a_x = -\frac{k}{m}x$$

- The magnitude of acceleration is proportional to the magnitude of the displacement of the block from equilibrium
- The direction of the acceleration is opposite to the position from equilibrium
- An object moves with simple harmonic motion whenever its acceleration is proportional to its position and is oppositely directed to the position from equilibrium

# Acceleration, final

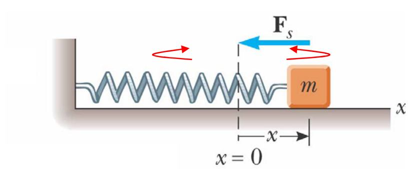



$$F_s = -kx$$
$$a_x = -\frac{k}{m}x$$

- The acceleration is ***not*** a constant (because the spring force keeps changing)
  - Therefore, the kinematic equations cannot be applied
  - If the block is released from some position $x = \boldsymbol{A}$, then the initial acceleration is $-\frac{k}{m}A$
  - When the block passes through the equilibrium position, the acceleration $a = 0$
  - The block continues to $x = -\boldsymbol{A}$ where its acceleration is $+\frac{k}{m}A$

9

# Motion of the Block

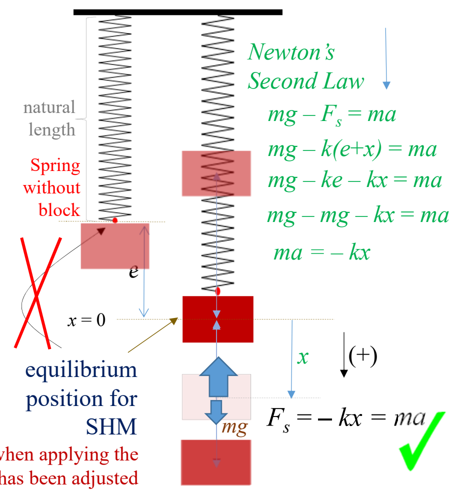



- The block continues to oscillate between $-\boldsymbol{A}$ and $+\boldsymbol{A}$
  - ✓These are turning points of the motion
- The force is conservative
- In the absence of friction, the motion will continue forever
  - ✓Real systems are generally subject to friction, so they do not actually oscillate forever

10

# What if it is a vertical spring!!

- When the block is hung from a vertical spring, its weight will cause the spring to stretch a distance $e$
- If the resting position of the spring (after the block is attached) is defined as $x = 0$, the same analysis as was done with the horizontal spring will apply to the vertical spring-mass system




The weight of the block needs not be considered when applying the Newton's second law if the equilibrium position has been adjusted away from the natural length to the resting position of the block after it is attached to the end of spring.

11

# Simple Harmonic Motion – Mathematical Representation

- Model the block as a particle
- Choose $x$ as the axis along which the oscillation occurs

$$\boxed{\begin{aligned} F_s &= -kx \\ a_x &= -\frac{k}{m}x \end{aligned}}$$

- Acceleration $a = \dfrac{d^2x}{dt^2} = -\dfrac{k}{m}x$
- We let $\omega^2 = \dfrac{k}{m}$, or $\omega = \sqrt{\dfrac{k}{m}}$

Angular frequency, or angular velocity

- Then $a = -\omega^2 x$

12

## Find the Mathematical Represe $a = \frac{d^2x}{dt^2} = -\omega^2 x$ HM

Angular frequency or angular velocity

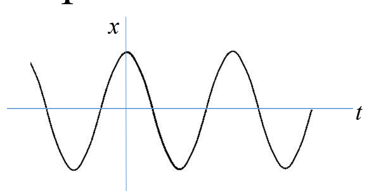
- A function of position $x$ with respect to time $t$ that satisfies the equation is needed
  - ✓Need a function $x(t)$ whose second derivative is the same as the original function with a negative sign and multiplied by $\omega^2$
  - ✓The sine and cosine functions meet these requirements

$$\sin(x) = \cos\left(x - \frac{\pi}{2}\right) = \cos\left(x + \frac{3\pi}{2}\right)$$

$$\cos(x) = \sin\left(x + \frac{\pi}{2}\right) = \sin\left(x - \frac{3\pi}{2}\right)$$

13

## Simple Harmonic Motion – Graphical Representation



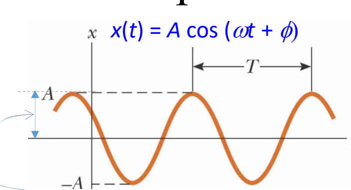

$$x = \cos\theta, \quad \frac{dx}{d\theta} = -\sin\theta$$

$$\frac{d^2x}{d\theta^2} = -\cos\theta$$

One of the generic solutions is

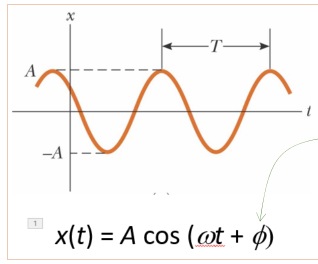
$x(t) = A\cos(\omega t + \phi)$ initial phase angle

where $A$, $\omega$, $\phi$ are all constants, and $t$ is the time in second




- $A$ is the amplitude of the motion. This is the maximum position of the particle in either the positive or negative direction
- $\omega$ is called the angular frequency (or angular velocity). Units of $\omega$ are rad/s
- $\phi$ is the phase constant or the initial phase angle

# Simple Harmonic Motion, cont

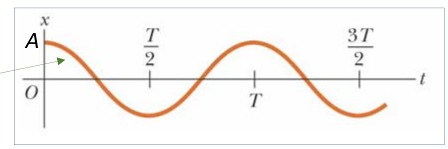



initial phase angle

- The ***phase*** of the motion is the quantity $(\omega t + \phi)$
- $x\,(t)$ is periodic and its value is the same each time $\omega t$ increases by $2\pi$ radians

  $\cos(\omega t + \phi) = \cos(\omega t + 2n\pi + \phi)$
- In the following diagram when $t = 0$, the particle is at $x = A$, then $\phi = 0$ as shown.

$x(t) = A \cos \omega t$

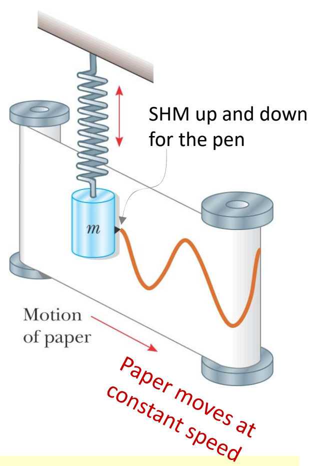


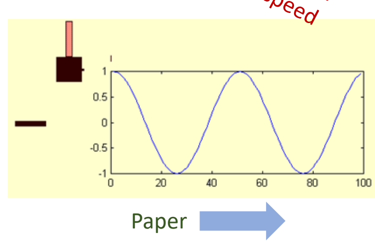

- For generic cases the initial phase angle $\phi$ may not be 0 when $t = 0$.

15

# An Experiment To Show SHM

- This is an experimental apparatus for demonstrating simple harmonic motion
- The pen attached to the oscillating object traces out a sinusoidal curve on the moving chart paper
- This verifies the cosine curve previously determined

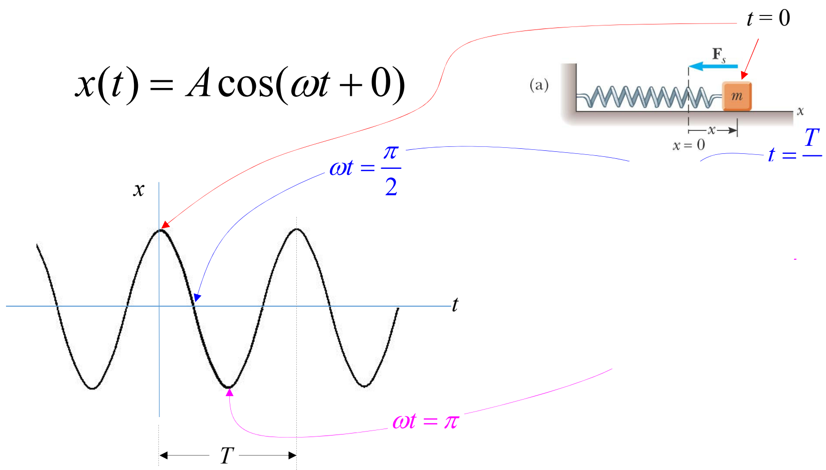



16

## Simple Harmonic Motion
## - The Cosine Function

$$x(t) = A\cos(\omega t + 0)$$




Period: Time required for 1 full cycle.



## Period $T = \dfrac{2\pi}{\omega}$

Regard the input to sine and cosine functions as angle

- $\omega = \dfrac{\theta}{t} = \dfrac{2\pi}{T}$ ➔ $T = \dfrac{2\pi}{\omega}$
- The ***period*** $T$, is the time interval required for the particle to go through one full cycle of its motion
  - The values of position *x* and velocity *v* for the particle at time *t* equal the values of *x* and *v* at time (*t* + $T$)

## Frequency $f = \dfrac{1}{T} = \dfrac{\omega}{2\pi}$

- The inverse of the period is called the ***frequency***
- The frequency represents the number of oscillations that the particle undergoes per unit time interval
- Units are cycles per second = hertz (Hz)

# Summary Equations – Period and Frequency

Algebraic Manipulation !!

$f = \frac{1}{T} = \frac{\omega}{2\pi}$

$\omega^2 = \frac{k}{m}$

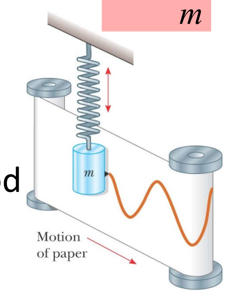
- The frequency and period equations can be rewritten to solve for $\omega$

$$f = \frac{\text{number of cycles}}{\text{time}}\text{, but } \omega = \frac{(\text{number of cycles}) \times 2\pi}{\text{time}}$$

$$\text{So } \omega = f \times 2\pi = 2\pi f = 2\pi\left(\frac{1}{T}\right) \rightarrow \omega = \frac{2\pi}{T}$$

- With $\omega = \sqrt{\frac{k}{m}}$, the frequency and period can also be expressed as:

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}\text{, and, } T = 2\pi\sqrt{\frac{m}{k}}$$



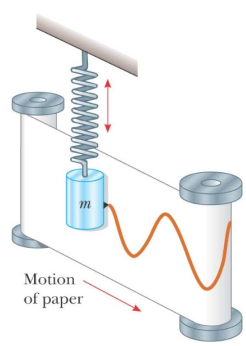



# Period and Frequency, cont

$$T = 2\pi\sqrt{\frac{m}{k}}$$

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

- The frequency and the period depend only on the mass of the particle and the force constant of the spring
- They do not depend on the parameters of motion (you can start with an initial velocity or you can pull the spring longer initially but these will have no effect on the period and frequency of the oscillation. Only amplitude is affected.)
- But if replaced by a stiffer spring (large values of $k$) and/or decrease the mass of the particle, frequency will be larger

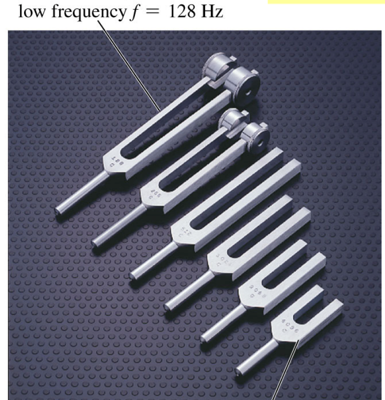


# Period and Frequency, cont

$$T = 2\pi\sqrt{\frac{m}{k}}$$

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

- The greater the mass $m$ in a tuning fork's tines, the lower the frequency of oscillation, and the lower the pitch of the sound that the tuning fork produces.






# Motion Equations for Simple Harmonic Motion

$\frac{dx}{dt}$ but $x$ is a function of $t$, i.e., $x = f(t)$

$$x(t) = A\cos(\omega t + \phi)$$

Then, $\frac{dx}{dt} = \frac{dx}{\quad} \times \frac{\quad}{dt}$

$$v = \frac{dx}{dt} = -\omega A \sin(\omega t + \phi)$$

$$\omega^2 = \frac{k}{m}$$

$$a = \frac{d^2x}{dt^2} = -\omega^2 A \cos(\omega t + \phi)$$

Acceleration is a variable!!

- Remember, simple harmonic motion is **not** uniformly accelerated motion. The magnitude and direction of acceleration in S.H.M. keeps changing.

# Maximum Values of $v$ and $a$

$$\omega^2 = \frac{k}{m}$$

$$x(t) = A\cos(\omega t + \phi)$$

$$v = \frac{dx}{dt} = -\omega A \cancelto{-1}{\sin(\omega t + \phi)}$$

$$a = \frac{d^2x}{dt^2} = -\omega^2 A \cancelto{-1}{\cos(\omega t + \phi)}$$

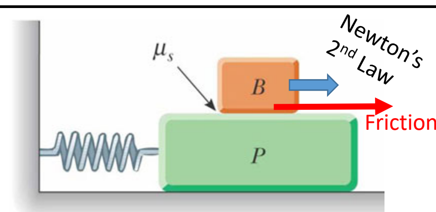
- Because the sine and cosine functions oscillate between $\pm 1$, we can find the maximum values of velocity and acceleration for an object in S.H.M.

$$v_{\max} = -\omega A \times (-1) = \sqrt{\frac{k}{m}}\, A$$

$$a_{\max} = -\omega^2 A \times (-1) = \frac{k}{m}\, A$$

23

**Example.** A large block P executes horizontal simple harmonic motion as it slides across a frictionless surface with a frequency $f$ = 1.50 Hz. Block $B$ rests on it as shown in figure, and the coefficient of static friction between the two is $\mu_s = 0.600$. What maximum amplitude of oscillation can the system have if block $B$ is not to slip? Assume that friction is negligible below Block $P$.




**Answer:**

Let the mass of Block $B$ be $m$. In order for Block $B$ to remain on Block $P$, the two blocks must have the same amplitude ($A$) of oscillation.

From $a = \dfrac{d^2x}{dt^2} = -\omega^2 A \cancelto{-1}{\cos(\omega t + \phi)}$, $\left(\dfrac{d^2x}{dt^2}\right)_{\max} = A\omega^2$.

"Friction will grip the bottom of Block B to perform SHM on Block B."

But $\text{Friction}_{\max} = \mu_s n = \mu_s mg$. So $m \times \left(\dfrac{d^2x}{dt^2}\right)_{\max} = \cancel{m} \times \left(A\omega^2\right) = \text{Friction}_{\max} = \mu_s \cancel{m} g$

$$\Rightarrow A = \frac{\mu_s g}{\omega^2} = \frac{\mu_s g}{(2\pi f)^2} = \frac{0.6 \times 9.8}{(2 \times 3.1416 \times 1.5)^2} = 0.066196 \text{ m} = \boxed{6.62 \text{ cm}}$$

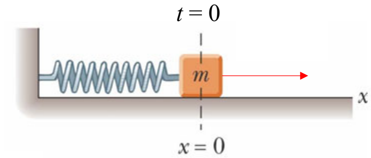
24

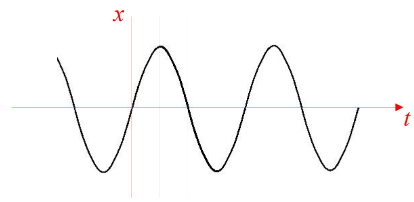
**Example.** A particle moving along the $x$ axis in simple harmonic motion starts from its equilibrium position (the origin) at $t = 0$ and moves to the right. The amplitude is 2.00 cm and the frequency is 1.50 Hz. Take going right as positive.

(a) Show that the position of the particle is given by

$x = (2.00\ \text{cm})\sin(3.00\pi t)$

**Answer:**




At $t = 0$, $x = 0$ and $v$ is positive (to the right) and the initial phase angle is 0. Therefore, this situation corresponds to $x = A\sin\omega t$ and $v = v_i\cos\omega t$.

Since $f$ = 1.50 Hz, $\omega = 2\pi f = 2\pi \times 1.50 = 3.00\pi$

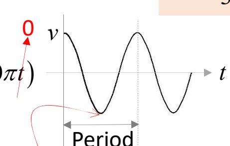
Also, $A$ = 2.00 cm, so $\boxed{x = (2.00\ \text{cm})\sin(3.00\pi t)}$

25

$\omega = 2\pi f$
$= 2\pi \times 1.50$
$= 3.00\pi$

(b) Determine the maximum speed and the earliest time ($t > 0$) at which the particle has this speed

**Answer:**



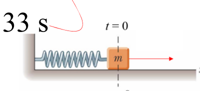

$$x = (2.00\ \text{cm})\sin(3.00\pi t) \rightarrow \frac{dx}{dt} = (2.00\ \text{cm})(3.00\pi)\cos(3.00\pi t)$$

$$v_{\max} = v_i = 2.00(3.00\pi)\times 1 = 6.00\pi\ \text{cm/s} = \boxed{18.8\ \text{cm/s}}$$

$T = \frac{1}{f} = \frac{1}{1.5} = 0.6667\ \text{s}$

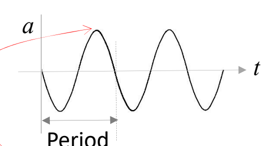
The particle has this speed at $t = 0$ (but this $t \not> 0$)

so the earliest time is at $t = \frac{T}{2} = 0.667 \times \frac{1}{2} = 0.3333\ \text{s}$




(c) Determine the maximum acceleration and the earliest time ($t > 0$) at which the particle has this acceleration

**Answer:**




$$\frac{d^2x}{dt^2} = -(2.00\ \text{cm})(3.00\pi)^2\sin(3.00\pi t)$$

$$a_{\max} = -2.00(3.00\,\pi)^2 \times (-1) = 18.0\,\pi^2\ \text{cm/s}^2 = \boxed{178\ \text{cm/s}^2}$$

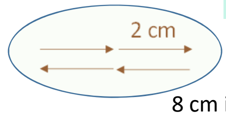
This positive value of acceleration first occurs at $t = \frac{T}{4} + \frac{T}{2} = \frac{3}{4}T = \boxed{0.500\ \text{s}}$

26

(d) Determine the total distance travelled between $t = 0$ s and $t = 1.00$ s.

$A = 2$ cm

$T = 0.667$ s




8 cm in 1 period of time

**Answer:**

For one cycle the period is 0.667 seconds and the amplitude is 2.00 cm. So the particle will travel (4x2.00 cm) = 8.00 cm in 0.667 seconds.

| Period | Time |
|---|---|
| 1 | 0.667 s |
| $x$ | 1 s |

$$x \times 0.667 = 1 \times 1 \;\rightarrow\; x = \frac{1}{0.667} = 1.5 \text{ Periods}$$

Hence between $t = 0$ s and $t = 1.00$ s, the particle will travel a distance of $8.00\text{ cm} \times 1.5 = \boxed{12.0\text{ cm}}$.

!!! You cannot use this ratio method to compute the distance travelled if the time duration is not a multiple of quarter-period because the velocity is not a constant so ratio cannot be used!!! In that case you will still need to use the sine function or cosine function.

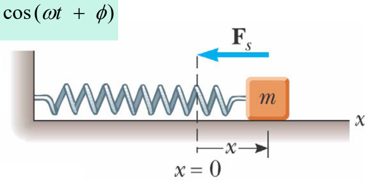
27

# Energy of the SHM Oscillator

$$x(t) = A\cos(\omega t + \phi)$$

$$v = \frac{dx}{dt} = -\omega A \sin(\omega t + \phi)$$

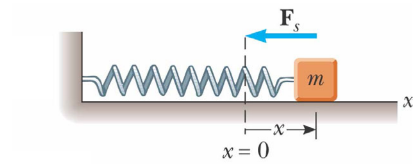
$$a = \frac{d^2x}{dt^2} = -\omega^2 A\cos(\omega t + \phi)$$



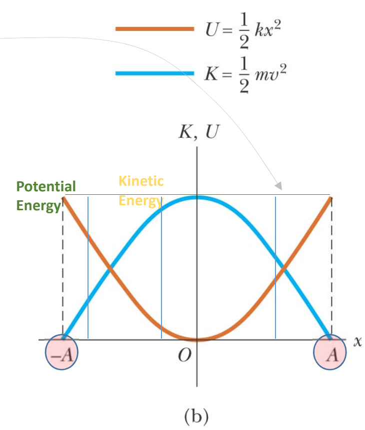

- Assume a spring-mass system is moving on a frictionless surface
- This tells us the total energy is constant

28

# Energy of the SHM Oscillator, cont



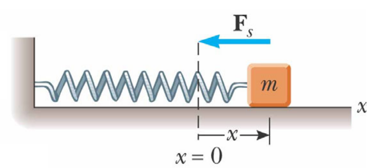

- The total mechanical energy is constant
- The total mechanical energy is proportional to the square of the amplitude




- Energy is continuously being transferred between potential energy stored in the spring and the kinetic energy of the block

29

# Energy of the SHM Oscillator, cont

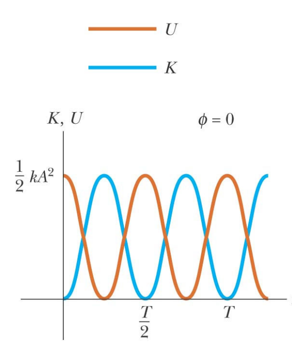



- As the motion continues, the exchange of energy also continues
- Energy can be used to find the velocity

$$\cancel{\frac{1}{2}}mv^2 + \cancel{\frac{1}{2}}kx^2 = \cancel{\frac{1}{2}}kA^2$$

$$\rightarrow v = \pm\sqrt{\frac{k}{m}\left(A^2 - x^2\right)}$$

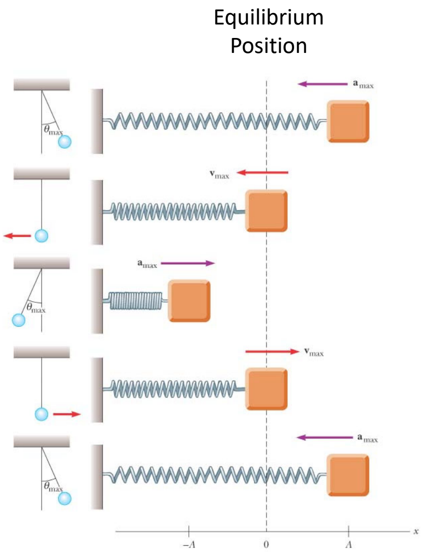
$$= \pm\omega\sqrt{A^2 - x^2}$$




$$\boxed{\omega^2 = \frac{k}{m}}$$

30

# Energy in SHM, summary




Kinetic Energy

Elastic Potential Energy

| $t$ | $x$ | $v$ | $a$ | $K$ | $U$ |
|---|---|---|---|---|---|
| 0 | $A$ | 0 | $-\omega^2 A$ | 0 | $\frac{1}{2}kA^2$ |
| $T/4$ | 0 | $-\omega A$ | 0 | $\frac{1}{2}kA^2$ | 0 |
| $T/2$ | $-A$ | 0 | $\omega^2 A$ | 0 | $\frac{1}{2}kA^2$ |
| $3T/4$ | 0 | $\omega A$ | 0 | $\frac{1}{2}kA^2$ | 0 |
| $T$ | $A$ | 0 | $-\omega^2 A$ | 0 | $\frac{1}{2}kA^2$ |

31

**Example.** A 65.0-kg bungee jumper steps off a bridge with a light bungee cord tied to herself and to the bridge. The unstretched length of the cord is 11.0 m. She reaches the bottom of her motion 36.0 m below the bridge before bouncing back. Her motion can be separated into an 11.0 m free fall and a 25.0 m section of simple harmonic oscillation. Assume that the air resistance is negligible.

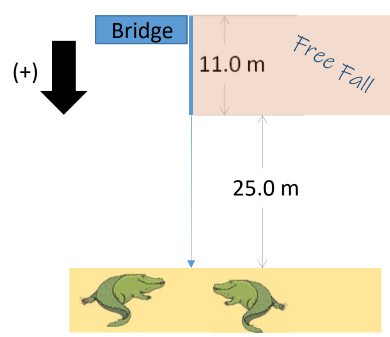
(a) For what time interval is she in free fall?

**Answer:**

$$s = s_0 + ut + \frac{1}{2}at^2$$

$$11 = 0 + 0 + \frac{1}{2}(9.8)t^2$$

$$\rightarrow t = \sqrt{\frac{22}{9.8}} = \boxed{1.50 \text{ s}}$$

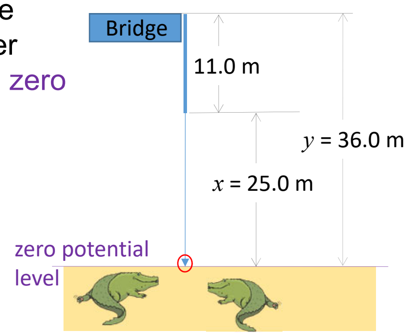



32

(b) Use the principle of conservation of energy to find the spring constant of the bungee cord.

**Answer:**

Take the initial point where she steps off the bridge and the final point at the bottom of her motion. Let the water surface shown in the figure as the zero potential level.




$$mgy = \frac{1}{2}kx^2$$

$$65 \times 9.8 \times 36 = \frac{1}{2}k(25)^2$$

$$\Rightarrow k = \boxed{73.4 \text{ N/m}}$$

33

(c) What is the location of the equilibrium point where the spring force balances the force of gravity acting on the jumper? What is the amplitude of the oscillation? Note that this point is taken as the origin in our mathematical description of simple harmonic oscillation.

**Answer:**

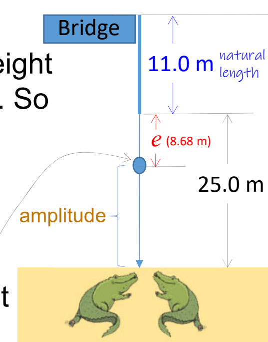
The equilibrium point is the point where the weight of the jumper rest motionlessly below the cord. So the tension on the cord has become $mg$.

Let the extension of the cord be $e$ due to $mg$.

From $F_s$ = - $ke$, $|-ke| = |mg|$ $\Rightarrow e = \frac{mg}{k} = \frac{65 \times 9.8}{73.4}$

$= 8.68$ m. So the location of the equilibrium point is $11 + 8.68 \text{ m} = \boxed{19.68 \text{ m below the bridge}}$

Amplitude of her oscillation is 36 – 19.68 = $\boxed{16.32 \text{ m}}$.




(d) What is the angular frequency of the oscillation?

**Answer:**

$$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{73.4}{65}} = \boxed{1.06 \text{ rad/s}}$$

34

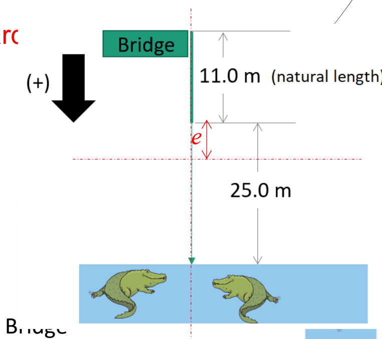
(e) What time interval is required for the cord to stretch by 25.0 m?

**Answer:** (regard the elastic cord as a horizontal spring!!)

We will approach this part in the reverse view with a re-defined $t = 0$.

Let $t$ = 0 at the maximum downwarc

From $x(t) = A\cos(\omega t + \phi),\ \phi = 0.$




So we need to find $t$ when
$x$ = - 8.68 m.

$$-8.68 = 16.32\cos\left(\overbrace{1.06t + 0}^{\text{radian}}\right)$$

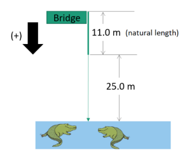
$$\rightarrow\ \cos(1.06t) = \frac{-8.68}{16.32} = -0.5319$$

$$\rightarrow\ \overbrace{1.06t}^{\text{radian}} = 122.13° = 122.13 \times \frac{\pi}{180} = 2.1316 \text{ rad. So } t = \frac{2.1316}{1.06} = \boxed{2.011 \text{ s}}$$

(f) What is the total time interval for the entire 36.0 m drop?

**Answer:**

Total time = $\overbrace{1.50}^{\text{Free fall}}\text{ s} + 2.011\text{ s} = \boxed{3.50\text{ s}}$

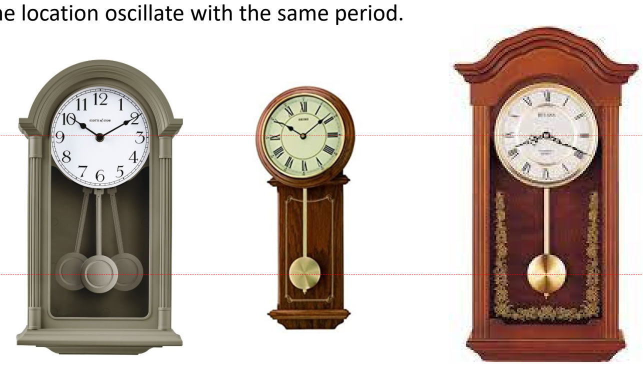



35

---

# Why?

All simple pendula that are of equal length and are at the same location oscillate with the same period.



The Maclaurin expansion of $\sin x$ is given by

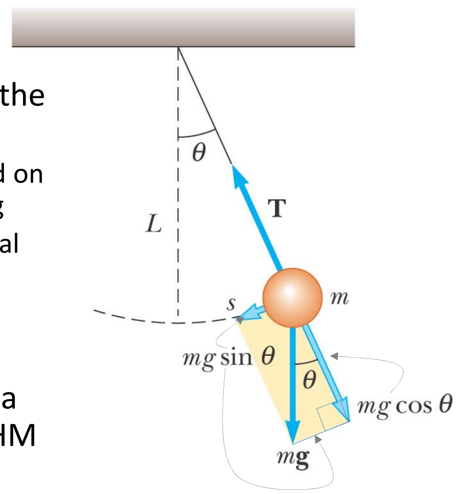
$$\sin(x) = \frac{x}{1!} - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots\ldots,$$ where $x$ is in radians.

When $x$ is small, we set $\sin(x) \approx x$. This is an approximation!!!

36

# Simple Pendulum

- The forces acting on the bob are **T** and *m***g**
  - **T** is the force exerted on the bob by the string
  - *m***g** is the gravitational force
- The tangential component of the gravitational force is a restoring force for SHM



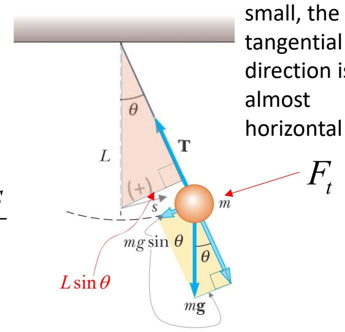

37

# Simple Pendulum

- Take going right to be positive.
- In the tangential direction,

$$F_t = -mg\sin\theta = ma = m\frac{d^2s}{dt^2}$$

When $\theta$ is very small, the tangential direction is almost horizontal!!




- The length, $L$, of the pendulum is constant.
- But for small values of $\theta$, $\sin\theta \approx \frac{s}{L}$, and, $\sin\theta \approx \theta$.

$$\text{So } ma = m\frac{d^2s}{dt^2} \approx m\frac{d^2(L\sin\theta)}{dt^2} = mL\frac{d^2(\sin\theta)}{dt^2} = mL\frac{d^2\theta}{dt^2}$$

$$F = ma \Rightarrow -\cancel{m}g\sin\theta = \cancel{m}L\frac{d^2\theta}{dt^2} \Rightarrow -g\theta \approx L\frac{d^2\theta}{dt^2} \Rightarrow \frac{d^2\theta}{dt^2} \approx -\frac{g}{L}\theta$$

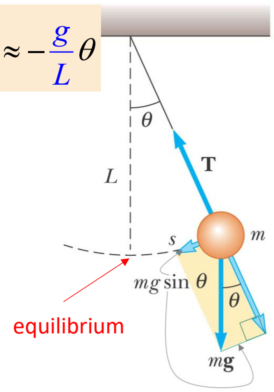
SHM only for small $\theta$

38

# Simple Pendulum

Let going right as positive. When time is 0, let the pendulum ball be at the extreme right end.

$$\frac{d^2\theta}{dt^2} \approx -\frac{g}{L}\theta$$




- The function $\theta$ can be written as

$$\theta = \theta_{max} \cos(\omega t + \phi)$$

- The angular frequency is $\omega = \sqrt{\frac{g}{L}}$
- The period is $T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{L}{g}}$
- The period and frequency of a simple pendulum depend only on the length of the string and the acceleration due to gravity
- The period is independent of the mass
- So all simple pendula that are of equal length and are at the same location oscillate with the same period



**Example.** The rise and fall of water in a harbour are simple harmonic. On a particular day in a harbour, high tide at its entrance occurs at 12 noon and the water depth is then 11.0 m, and low tide occurs 6.25 hours later and the water depth is then 5.0 m.

(i) Find the amplitude and period of this motion.

(ii) Find the time when the water level will be falling at its maximum rate and thus find the rate in meter per minute.

(iii) A ship needs a depth of 7.0 m to enter the harbour. Find the latest time after noon at which it can enter without having to wait for low tide to pass.

(iv) If the ship arrived at 7.00 p.m., what is the next possible time it can enter the harbour?

Answer:

(i) Amplitude $A = \frac{11.0 - 5.0}{2} = 3\text{ m}$

Period (time for high tide to low tide, and back to high tide)

$T$ = 6.25 × 2 = 12.5 hours

## Answer:

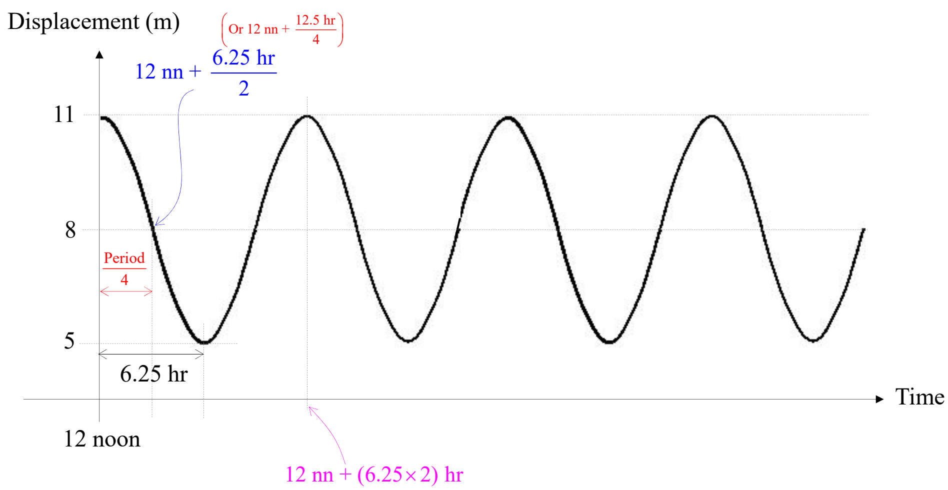
Period = 12.5 hours

The rise and fall of water in a harbour are simple harmonic. On a particular day in a harbour, high tide at its entrance occurs at 12 noon and the water depth is then 11.0 m, and low tide occurs 6.25 hours later and the water depth is then 5.0 m.

(i) Find the amplitude and period of this motion.

(ii) Find the time when the water level will be falling at its maximum rate and thus find the rate in meter per minute.

(ii) At the equilibrium level, speed of water is a maximum.




41

## Answer:

Period = 12.5 hours

The rise and fall of water in a harbour are simple harmonic. On a particular day in a harbour, high tide at its entrance occurs at 12 noon and the water depth is then 11.0 m, and low tide occurs 6.25 hours later and the water depth is then 5.0 m.

(i) Find the amplitude and period of this motion.

(ii) Find the time when the water level will be falling at its maximum rate and thus find the rate in meter per minute.

(ii) At the equilibrium level, speed of water is a maximum.

Required time measured from noon = $\dfrac{\text{period}}{4} = \dfrac{12.5}{4} = 3.125$ hour

$0.125 \times 60$ minutes $= 7.5$ minutes

Hence, the time when the water level will be falling at its maximum rate is at 12 noon + 3 hr and 7.5 min = 3:07:30 p.m.

$$\omega = \frac{\theta}{t} = \frac{2\pi}{T}$$

$$x(t) = A\cos(\omega t + \overset{0}{\phi}) \rightarrow v(t) = -A\omega \overset{1}{\sin}(\omega t) \qquad \boxed{\omega t = \frac{\pi}{2}}$$

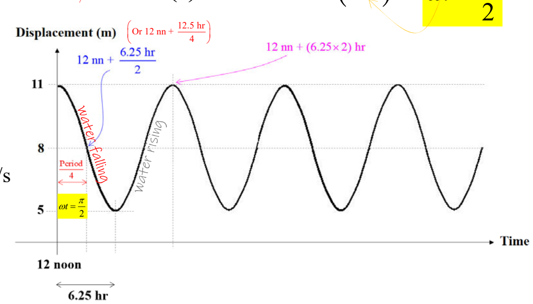
Maximum water falling rate:

$$v_{max} = A\omega = A \times \left(\frac{2\pi}{T}\right)$$

$$= 3.0\left(\frac{2\pi}{12.5 \times 3600}\right) = 4.189 \times 10^{-4} \text{ m/s}$$

$$= \left(4.189 \times 10^{-4}\right) \times 60 \text{ m/min}$$

$$= 0.0251 \text{ m/min}$$




42

(iii) A ship needs a depth of 7.0 m to enter the harbour. Find the latest time after noon at which it can enter without having to wait for low tide to pass.
(iv) If the ship arrived at 7.00 p.m., what is the next possible time it can enter the harbour?

## Answer:

(iii) $x(t) = A\cos(\omega t + \phi) = A\cos\left(\left(\frac{2\pi}{T}\right)t + 0^\circ\right)$

$$\boxed{-1.0 = 3\cos\left(\left(\frac{2\pi}{12.5\times 3600}\right)t\right)}$$

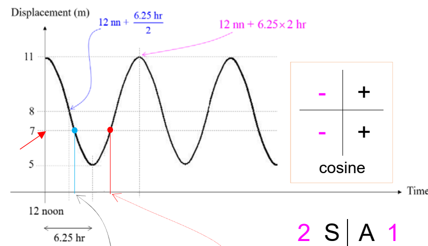
(2nd Quadrant)

$$\Rightarrow \frac{2\cancel{\pi}t}{12.5\times 3600} = \cos^{-1}(-0.3333) = 180^\circ - 70.529^\circ$$

$$= 109.471^\circ = 109.471^\circ \times \frac{\pi}{180^\circ}\ \text{rad} = 0.6082\cancel{\pi}$$

$$\Rightarrow t = \frac{0.6082\times 12.5\times 3600}{2} = 13683.875\ \text{s}$$

$$= \frac{13683.875}{3600}\ \text{hr} = 3.8\ \text{hr} \quad (0.8\ \text{hr}\times 60 = 48\ \text{min})$$




| 2 S | A 1 |
|---|---|
| 3 T | C 4 |

Hence, the latest time after 12 noon at which the ship can enter without having to wait for low tide to pass is 3:48p.m.

(iv) Another solution for $\boxed{-1.0 = 3\cos\left(\frac{2\pi t}{12.5\times 3600}\right)}$

(3rd Quadrant)

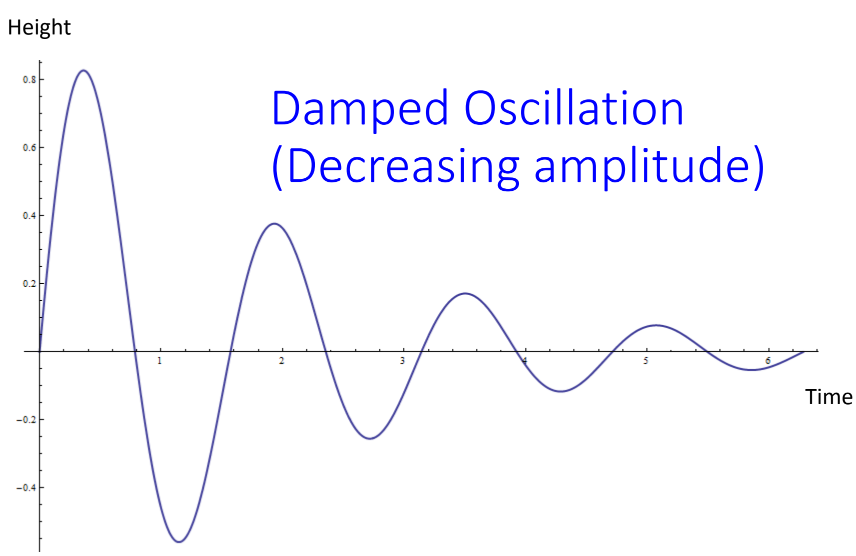
$$\frac{2\cancel{\pi}t}{12.5\times 3600} = 180^\circ + 70.529^\circ = 250.529^\circ = 250.529^\circ \times \frac{\pi}{180^\circ}\ \text{rad} = 1.3918\cancel{\pi}$$

$$\Rightarrow t = \frac{1.3918\times 12.5\times 3600}{2} = 31316.125\ \text{s} = \frac{31316.125}{3600}\ \text{hr} = 8.6989\ \text{hr}$$

$$(0.6989\ \text{hr}\times 60 = 41.934\ \text{min})$$

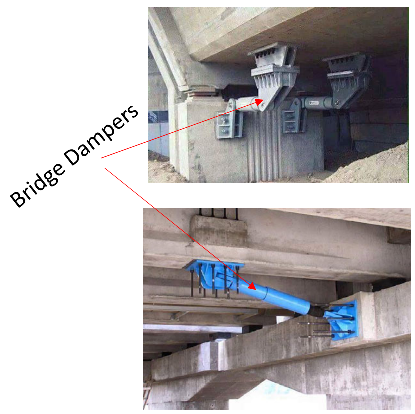
Hence, the next possible time the ship can enter the harbour is 8:42 p.m.

43




44

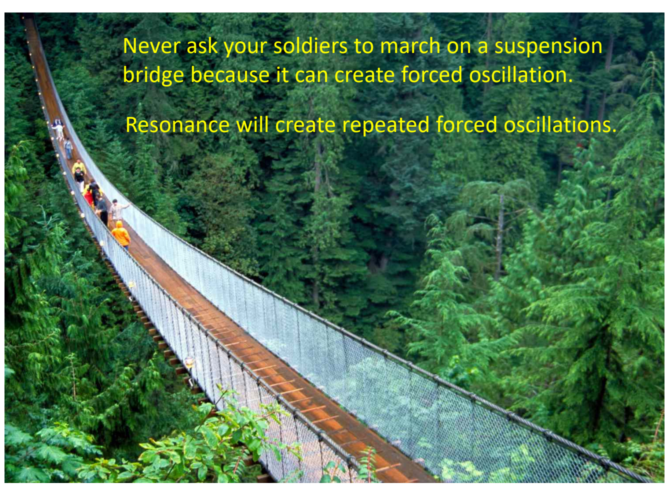
# Importance of the Use of Dampers in Civil Engineering




The purpose of damper is to absorb (deplete) the energy due to vibration.

It is used to mitigate vibration due to
- Strong wind
- Earthquake
- High human traffic

Power delivered by resonance is tremendous and this can cause a bridge to collapse.

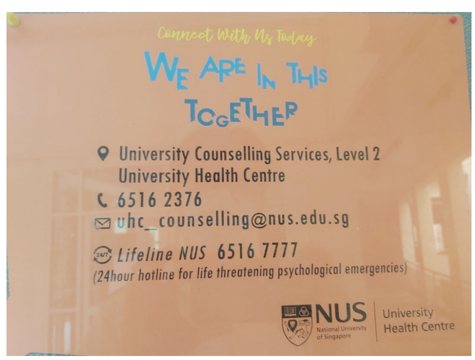


Forced Oscillations (Just for your knowledge.
This is not in the syllabus of PC1201 )
Video credit: https://www.youtube.com/watch?v=wb4Q-eOvyBg




**Helplines to de-stress**

We are not superman/superlady. Sometime and somewhere in our life journey we will need help. Here they are:




Lifeline NUS (for life threatening psychological emergencies):
6516 7777 (24 hours)

6516 2376 (Office hours)
uhc_counselling@nus.edu.sg

Samaritans of Singapore Hotline: 1800 221 4444
Institute of Mental Health's Helpline: 6389 2222
Singapore Association of Mental Health Helpline: 1800 283 7019